

Subek Acharya

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Summary

Computer Engineer and AI Researcher specializing in adversarial machine learning and AI security with experience in training and evaluating ML models, building production-grade ETL pipelines, and deploying AI systems. Skilled problem solver with a proven ability to work independently and collaboratively across engineering, research, and academic environments to deliver impactful solutions.

Work Experience

Graduate Research Assistant

Jan 2025 – Present

RSAIL, University of Rhode Island

Kingston, RI

- Collaborated with the Voting Technology Center at the University of Connecticut on adversarial security research in the election domain
- Conducted large-scale experiments on the Unity HPC Platform, managing distributed training jobs using the SLURM workload manager across multi-node GPU clusters
- Trained and benchmarked diverse model architectures including Residual Networks, VGG Networks, Class-Attention Transformers, Support Vector Machines, Spiking Neural Networks, Prototype Networks, State Space Models, and Vision Transformers on UConn Bubbles ballot and MVTec/VisA industrial defect datasets
- Developed *Project Black*: a multi-norm black-box adversarial attack framework benchmarking 11 black-box and five transfer-based white-box attacks across ℓ_∞ , ℓ_2 , ℓ_1 , and ℓ_0 perturbation norms on digital and physical threat domains
- Curated an adversarial evaluation dataset of 32,000 printed and scanned adversarial ballot examples to evaluate real-world physical attack effectiveness
- Authored 3 research papers (1 first-authored accepted to IEEE PAINE 2026, 2 under review) on adversarial robustness of AI systems in election and industrial defect detection domains

Graduate Assistant

Jun. 2025 – Present

ITS Teaching and Learning Services, University of Rhode Island

Kingston, RI

- Automated TLS Data Report generation using OpenCode with Claude Opus 4.7 inference, leveraging Python's Matplotlib library and LaTeX to produce structured, publication-ready reports
- Developed custom Brightspace Adoption Dashboard by engineering an ETL pipeline to ingest D2L CSV exports into a SQL database and visualizing platform adoption metrics in Power BI for IT leadership, delivering analytics capabilities equivalent to Brightspace Performance+ tier and saving \$50,000+/year in licensing costs
- Led technical implementation and platform management of Webex Events (Socio) for virtual seminars Tech Day 2025 & Tech Day 2026, onboarding the platform for the first time at URI with 150+ participants from 15+ institutions on each event
- Provided technical support to faculty on Brightspace LMS by resolving service requests through ServiceNow ticketing system

Graduate Assistant

Oct. 2024 – Jan. 2025

URI Health Services, University of Rhode Island

Kingston, RI

- Configured and deployed online appointment system within the Mediat Patient Portal (college health EHR platform), shifting 80% of bookings to digital and reducing triage staff scheduling workload by an estimated 3 hours/day
- Developed Financial Payment Distribution Dashboard using Mediat BI (Tableau-integrated platform), replacing manual report generation and enabling real-time visualization of health charges and payment data of 12,000+ URI students for the Health Services administrative department

Co-Founder and CEO

Nov. 2020 – Jun. 2024

Wonder Groups Pvt. Ltd.

Kathmandu, Nepal

- Led implementation of School ERP Systems and network infrastructure for 12 government schools across Bagmati Province, Nepal under the President Educational Reform Program
- Managed a cross-functional team of 4 engineers, coordinated with government stakeholders, and conducted end-user training for 100+ faculty and administrative staffs

Software Engineer

Jun. 2018 – Sep. 2020

Cotiviti Nepal Pvt. Ltd.

Kathmandu, Nepal

- Designed and maintained 10+ ETL pipelines using Oracle Data Integrator (ODI) to optimize data integration workflows processing 10M+ US healthcare insurance records
- Developed PL/SQL and Python scripts to automate extraction, transformation, and loading of healthcare claims, RX, and eligibility data
- Designed and managed data schemas in Oracle DB for 5+ clients, ensuring data cleanliness, field mapping accuracy, and HIPAA compliance for protected health information (PHI)

 Technical Skills

AI & Machine Learning: PyTorch, Hugging Face Transformers, LoRA fine-tuning, vllm, LangChain, LangGraph, Distributed Training on SLURM, Weights & Biases
Integration & ETL: REST APIs, Web Services, FastAPI, Oracle Data Integrator
Programming Languages: Python,C/C++, PL/SQL, Java, JavaScript, Bash Scripting
Databases: Oracle Database, SQL Server, MongoDB
Operating Systems: Linux/Unix, Windows Server
Tools & Platforms: Git/GitHub, Docker, Kubernetes, PowerBI, Tableau, GitHub Copilot, Claude Code, AWS Kiro

 Projects

- Training & Evaluation of PyTorch models on CIFAR-10** | [GitHub](#)
- Trained CNN models (ResNet-18, VGG-16, DenseNet-121, GoogLeNet) on CIFAR-10 dataset and evaluated clean accuracy and adversarial robustness performance against benchmark attacks
- Training and Evaluation of Spiking Neural Network models on CIFAR-10** | [GitHub](#)
- Implemented SNN VGG-16 and SNN ResNet-20 architecture for CIFAR-10 classification using SpikingJelly framework
- Black-Box Adversarial Attack Framework** | [GitHub](#)
- Implemented a comprehensive black-box adversarial attack framework in PyTorch with four attack methods (RayS, ADBA, Square, SurFree)
- Adaptive Black-Box Adversarial Attack Framework** | [GitHub](#)
- Developed a black-box adversarial attack framework that trains a synthetic substitute model using oracle-labeled data, then executes six white-box attacks across multiple norms to generate transferable adversarial examples
- Training and Evaluation of Explainable AI Models** | [GitHub](#)
- Developed interpretable deep learning models using ProtoPNet architecture with ResNet-20 and VGG-16 backbones
- RAG Application: Document Q&A System** | [GitHub](#)
- Built a document question-answering system using the LangChain framework with Mistral-7B, ChromaDB vector store, and Cohere reranking, providing answers with source citations and confidence scores

 Publications

- PatchGuard Lacks Robustness: Breaking State-Of-The-Art Defect Detection Under New Norm Attacks.**
Subek Acharya, Syed Ayon, Khaleque Md Aashiq Kamal, Ruyu Zhou, Kaleel Mahmood. (*IEEE PAINE 2026*)
- Black-Box Adversarial Example Attacks on Machine Learning Based Election Systems.**
Subek Acharya, Syed Ayon, Khaleque Md Aashiq Kamal, Alina Barnett, Surya Eada, Kaleel Mahmood. (Preprint)
- Analyzing Physical Adversarial Example Threats to Machine Learning in Election Systems.**
Khaleque Md Aashiq Kamal, Surya Eada, Aayushi Verma, **Subek Acharya**, Adrian Yemin, Benjamin Fuller, Kaleel Mahmood. (Preprint)

 Education

University of Rhode Island <i>Master of Science in Computer Science, GPA: 3.93</i>	Sep. 2024 – Dec. 2026 Kingston, RI
Relevant Coursework: Machine Learning, Neural Networks and Deep Learning, Topics in Large Language Models, Machine Learning Security & Privacy, Machine Learning Evaluation and Explanation	
Tribhuvan University <i>Bachelor of Engineering in Computer Engineering</i>	Dec. 2013 – Oct. 2017 Lalitpur, Nepal
Relevant Coursework: Software Engineering, Operating Systems, Computer Networks, Data Structures and Algorithms, Database Management Systems	

 Leadership

Executive Council Member Webmaster <i>Graduate Student Association, URI</i>	Jun. 2025 – May 2026
Coordinator <i>Global Connections, URI</i>	Sep. 2024 – Dec. 2025